

CHERNYAEVO, Russian Federation

WMO: 313710

Lat: 52.7896N

Lon: 126.0030E

Elev: 209

StdP: 98.84

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.1	-37.7	-42.5	0.1	-39.9	-40.2	0.1	-37.5	8.7	-17.7	7.7	-20.0	0.1	50	0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	30.5	20.6	28.6	19.7	26.8	19.0	22.3	27.5	21.3	26.2	20.3	24.8	2.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	15.8	24.7	19.6	14.7	23.4	18.7	13.9	22.4	67.1	27.7	63.1	26.2	59.5	24.8	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.0	6.9	6.0	-42.5	33.9	2.6	2.1	-44.3	35.4	-45.8	36.6	-47.3	37.8	-49.2	39.3
			-42.4	23.9	2.4	1.3	-44.2	24.9	-45.6	25.6	-46.9	26.4	-48.7	27.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.2	-25.3	-20.5	-10.4	2.1	10.6	17.7	20.6	18.0	10.5	0.1	-14.8	-24.5
	DBStd	17.24	6.24	5.59	6.41	4.46	3.93	3.47	2.81	3.14	4.17	4.99	6.96	6.02
	HDD10.0	5026	1096	855	631	239	40	1	0	1	43	307	743	1071
	HDD18.3	7286	1354	1088	889	487	242	51	11	42	236	564	993	1329
	CDD10.0	932	0	0	0	3	58	233	328	250	59	1	0	0
	CDD18.3	149	0	0	0	0	2	33	80	33	1	0	0	0
	CDH23.3	1490	0	0	0	2	52	439	669	307	20	0	0	0
	CDH26.7	418	0	0	0	0	10	130	206	71	1	0	0	0
Wind	WSAvg	2.0	1.7	1.8	2.4	2.7	2.6	1.9	1.7	1.8	2.2	2.3	1.9	1.8
Precipitation	PrecAvg	485	6	5	9	31	43	73	121	93	65	19	13	8
	PrecMax	762	26	20	24	84	126	175	210	235	158	57	33	22
	PrecMin	296	0	0	1	7	8	19	22	34	12	0	2	1
	PrecStd	91	6	5	6	16	27	42	45	51	36	15	8	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.7	-2.5	9.1	21.8	28.3	32.9	33.7	31.1	25.7	17.4	2.6	-5.7
		MCWB	-9.5	-5.1	3.0	11.2	16.7	22.3	21.5	21.1	17.5	10.3	-0.6	-7.7
	2%	DB	-11.0	-6.1	5.5	16.7	24.5	29.9	30.9	28.6	23.0	13.5	-0.3	-9.8
		MCWB	-12.3	-7.7	0.6	8.2	13.9	19.5	21.2	20.3	16.0	7.8	-2.3	-10.9
	5%	DB	-13.3	-8.6	2.8	13.4	21.8	27.9	28.9	26.6	20.5	10.6	-2.3	-13.2
		MCWB	-14.3	-10.2	-1.2	6.1	12.7	18.5	20.3	19.4	14.4	5.7	-3.7	-14.2
	10%	DB	-15.6	-11.0	0.4	10.7	19.2	25.9	27.1	24.7	18.2	8.0	-4.6	-15.5
		MCWB	-16.4	-12.4	-2.6	4.4	11.2	17.7	19.8	18.6	12.9	3.7	-6.1	-16.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.3	-5.1	3.4	11.3	17.3	23.7	24.2	22.8	18.7	11.2	-0.1	-7.4
		MCDB	-7.7	-3.0	9.0	21.0	26.0	29.0	29.5	27.8	23.6	16.5	1.7	-5.8
	2%	WB	-12.3	-7.7	0.8	8.4	15.1	21.1	22.7	21.5	16.9	8.1	-2.0	-10.9
		MCDB	-11.0	-6.0	4.9	15.7	22.7	28.0	27.6	26.7	21.6	13.2	-0.4	-9.8
	5%	WB	-14.3	-10.1	-0.9	6.4	13.5	19.6	21.8	20.4	15.6	6.0	-3.8	-14.2
		MCDB	-13.3	-8.6	2.5	12.9	19.7	25.7	26.5	24.8	19.2	9.6	-2.3	-13.3
	10%	WB	-16.4	-12.3	-2.7	4.7	12.1	18.3	20.8	19.4	14.1	4.3	-6.1	-16.3
		MCDB	-15.6	-11.0	0.3	9.9	17.9	24.2	25.3	23.3	17.2	7.7	-4.7	-15.5
Mean Daily Temperature Range	5% DB	MDBR	11.5	15.0	14.8	12.7	12.9	12.5	10.6	11.2	12.2	10.8	10.8	10.3
		MCDBR	10.9	16.2	15.4	18.3	18.6	16.4	14.7	14.6	15.9	15.4	11.6	9.9
		MCWBR	9.9	14.3	11.1	10.8	9.5	7.2	6.1	7.0	8.7	10.0	9.4	9.1
	5% WB	MCDBR	10.7	16.2	14.7	17.3	15.6	13.8	11.3	11.8	13.2	14.2	11.5	9.8
		MCWBR	9.8	14.5	10.8	10.4	8.4	6.8	5.5	6.2	7.8	9.5	9.5	9.0
Clear-Sky Solar Irradiance	taub	0.260	0.293	0.360	0.535	0.593	0.524	0.484	0.421	0.387	0.369	0.303	0.265	
	taud	2.144	2.086	1.955	1.700	1.670	1.932	2.103	2.274	2.283	2.162	2.129	2.084	
	Ebn at Noon	717	799	802	692	677	741	767	801	781	694	638	619	
	Edh at Noon	69	100	143	212	230	178	148	118	104	94	70	61	
All-Sky Solar Radiation	RadAvg	1.35	2.52	4.24	5.28	5.31	6.07	5.31	4.70	3.62	2.35	1.45	1.03	
	RadStd	0.05	0.07	0.16	0.50	0.39	0.61	0.51	0.36	0.28	0.16	0.07	0.05	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page